

Speed of Light



The speed of light is found using a 1335p HeNe laser, two photodiode detectors, and a fast oscilloscope. The laser beam is split and time difference for pulses to reach detector at various distances are measured. The speed of light is found to be $c = (3.01 \pm 0.04) \times 10^8 m/s$. This result is within one standard deviation of the known value. The longitudinal mode spacing for our laser tube is determined by measuring the period between pulses being detected. The longitudinal mode spacing is determined to be $f = (0.33 \pm 0.02) GHz$. This result is within one standard deviation of the known value.

I. INTRODUCTION

The speed of light is a fundamental constant in the universe. It is known to be 2.998×10^8 meters per second in a vacuum[1], and negligibly slower in air. Like anything with a speed, light-speed can be measured by finding the amount of time that it takes light to travel a known distance. We can use Eq.1 to find the speed of light.

$$c = \frac{\Delta x}{\Delta t} \quad (1)$$

Eq.1 Speed formula with c as the speed of light, Δx as the known distance traveled, and Δt as the time taken to travel. Eq.1 is also the slope of a linear x vs t graph.

An obvious problem arises as to how we can practically measure the immense magnitude of light-speed for a photon. The method that we have employed for this experiment is that of exploiting the fact that a 1335p HeNe laser naturally 'lases' at different modes with different frequencies. The lasing occurs for modes with frequencies determined by Eq.2.

$$f_n = n \frac{c}{2L} \quad (2)$$

Eq.2 Mode frequencies for the HeNe laser.[2] f_n is the mode frequency, n is an integer determined by the mode and length of the laser tube, c is the speed of light, and L is the length of the laser cavity tube.

The length of our laser tube allows many modes near the principle mode to occur at once (i.e. modes occurring at $n \pm 1$, $n \pm 2$, etc.). The presence of different frequencies associated with these modes form a beat; these beats within the beam exist together to form pulses. Because the emitted light is not truly continuous, we can split the laser beam and emplace fast photodiode detectors at two different points (Δx) to find the difference in time (Δt) between a pulse making contact with the first and second detectors.

The modes that contribute most to the laser pulse are consecutive to the principle mode and make up the bulk of the peak on the scope. The beat frequency, or

longitudinal mode spacing is the frequency difference between the principle modes and is determined by Eq.3.

$$f = \frac{c}{2L} \quad (3)$$

Eq.3 Beat frequency of the principle modes of the HeNe laser.[2]

II. EXPERIMENTAL PROCEDURE

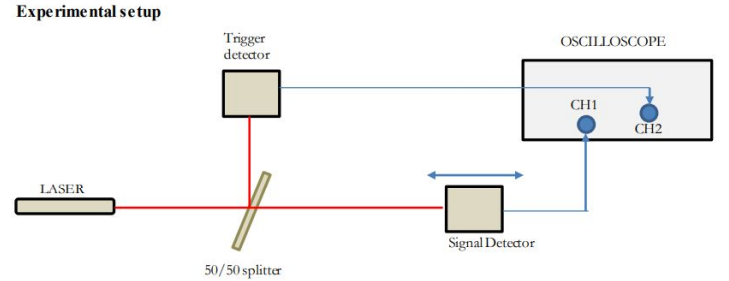


Fig.1 Block Diagram for this speed of light experiment.[3] Shows the HeNe laser, 50% beam splitter, trigger and signal detectors, and oscilloscope.

To set up this experiment, we shine the light of a HeNe laser through a 50% beam splitter. Half of the beam goes into a fixed trigger detector whose purpose is to detect the presence of the laser. It prompts the oscilloscope to begin taking data when the rising edge of the light intensity (front of the pulse) enters the detector.

A second, movable signal detector is placed at a starting distance from the beam splitter. Its purpose is to detect pulses from the laser at varying distances that we position the detector.

We start by placing the signal detector at some distance from the beam splitter and note the time on the oscilloscope of the leftmost peak. Part of our experiment involves calculating the longitudinal mode spacing, so before we move the detector, we measure the period between consecutive peaks on the screen. We then move the signal detector a small distance and note the new time position on the scope of the same pulse; we repeat this for the length of the table. The goal should be to

'follow' a certain peak as it moves across the screen when we move the detector. The difference in time that results from moving the signal detector further out is the extra time that the light must travel to reach the new position.

It is important to move the detector less than about 30cm each measurement. This is because the speed of light is about 30cm/ns. The time scale on our oscilloscope is in the nanosecond range, so moving the detector 30cm or more will make it difficult to determine which peak on the screen is the one we are meant to be following.

The speed of light can be calculated from the slope of a linear graph of detector position versus time position of the peak we follow.

The longitudinal mode spacing is also known as the beat frequency and can be found as the inverse of the peak period that we find before moving our detector.

III. ANALYSIS OF DATA

We moved the signal detector in regular intervals of 25cm to record the following data:

Table 1 Time position (t) of signal pulses for a given signal detector position (x), and their uncertainties.

n	$t \pm 0.2$ (ns)	$x \pm 0.25$ (cm)
0	-0.48000	11.25
1	0.26000	36.25
2	1.1400	61.25
3	1.9200	86.25
4	2.8800	111.25
5	3.5800	136.25
6	4.4800	161.25
7	5.3000	186.25
8	6.1400	211.25
9	6.9600	236.25
10	7.7600	261.25
11	8.6400	286.25
12	9.4200	311.25
13	10.320	336.25
14	11.120	361.25
15	12.020	386.25
16	12.740	411.25
17	13.580	436.25
18	14.500	461.25
19	15.240	486.25

The uncertainty in position is a quarter centimeter at every position to account for imperfections in our measurement setup and procedure. The time uncertainty is the largest observed drift in the time position on the oscilloscope when remeasuring a few of our points after taking our initial set of data. This difference is caused by beam instability from thermal cycling processes that cause laser gain curve shifts to occur within the laser.

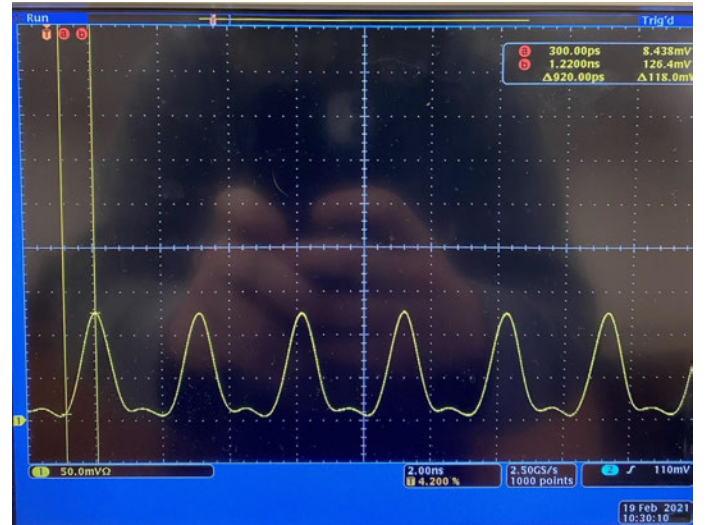


Fig.2 Example of a measurement taken on an oscilloscope. NOTICE: This image was from Phil Dyche's separate data taking session. The flash drive brought to this report's session corrupted and failed to save the images from our own oscilloscope.

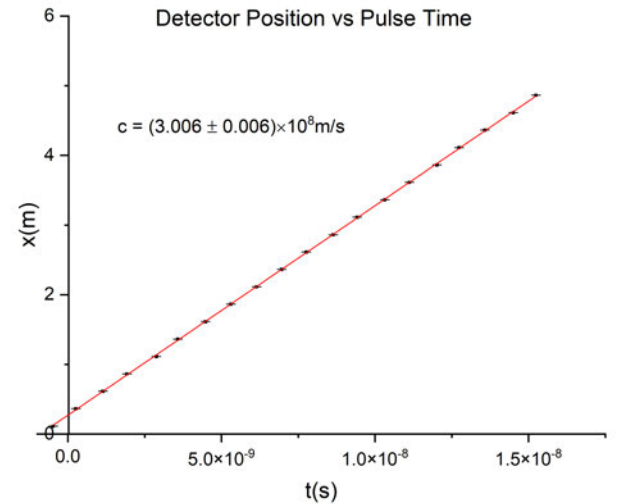


Fig.3 Detector position plotted against time position of signal pulses that enter the detector at various positions. The slope of Fig.1 gives a speed of light of $c = (3.006 \pm 0.006) \times 10^8 \text{ m/s}$.

To ensure an accurate result to cover the known speed of light with a single error value, we broaden our final uncertainty using a rough uncertainty method. We find the difference between the highest possible and lowest possible slopes according to the endpoints of our data and their uncertainties. We then divide by the square root of the number of measurements that we took. Doing this gives us a result of:

$$c = (3.01 \pm 0.04) \times 10^8 \text{ m/s} \quad (4)$$

The period that we measured between consecutive peaks

on our oscilloscope is $3.050ns \pm 0.2ns$. The uncertainty is from the same drift caused by the thermal cycling instability.

We now find the longitudinal mode spacing. We may find the difference between the highest contributing mode frequencies by equating Eq.3 to the inverse of the period between pulses on the oscilloscope.

$$f = \frac{1}{T} = \frac{c}{2L} \quad (5)$$

Eq.4 Longitudinal mode spacing, or beat frequency for the 1335p HeNe laser. T is the period between pulses.[3]

The final uncertainty for the beat frequency is found by taking the partial derivative of Eq.4 with respect to T and inserting our $0.2ns$ drifting uncertainty. This all gives results of:

$$f = (0.33 \pm 0.02)GHz \quad (6)$$

IV. DISCUSSION AND CONCLUSION

Our speed of light was determined to be $c = (3.01 \pm 0.04) \times 10^8 m/s$. This is within one standard deviation of the theoretical value of $2.998 \times 10^8 m/s$. The relative uncertainty is about 1.3%. Our final result is both accurate and precise.

Our longitudinal mode spacing for our 1335p HeNe laser was determined to be $f = (0.33 \pm 0.02)GHz$. This is within one standard deviation of the known value of $0.320GHz$. [2] The relative uncertainty is about 6%. This result is accurate and precise.

In these experiments we were able to utilize a natural phenomenon of lasers to find an accurate speed of light. We were also able to find the longitudinal mode spacing, or the beat frequency, the spacing between active 'lasing' modes that occur naturally within the laser tube.

[1] R. Penrose, *The Road to Reality: A Complete Guide to the Laws of the Universe*, Vintage Series (Vintage Books, 2007).

[2] T. Colbert, Colbert spots speed of light additional comments, [REDACTED]

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(2021), accessed on 2021-02-16.

[3] T. Colbert, Colbert spots speed of light task sheet, [REDACTED]

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